

Stats101A_Group4_Final_Project

Group4

2026-08-03

Data cleaning and descriptive report

```
## -----
## 1. Read raw merged data + basic QC
## -----
data <- read.csv("rentcast_ucla_rentals_15zips_full.csv")

# Confirm the zipCode RentCast returned matches the zip code we searched for
mismatch <- data %>% filter(as.character(zipCode) != as.character(search_zip))
nrow(mismatch)

## [1] 0

table(mismatch$search_zip, mismatch$zipCode)

## < table of extent 0 x 0 >

## -----
## 2. Variable selection
## -----
# Keep the 6 core variables + zipCode, plus latitude/longitude
# (kept temporarily to test as an alternative to zipCode -- see Step 4)
data_clean <- data %>%
  select(price, zipCode, bedrooms, bathrooms, squareFootage,
         propertyType, yearBuilt, latitude, longitude)

str(data_clean)

## 'data.frame': 2519 obs. of 9 variables:
## $ price : int 2795 2400 7500 7295 2500 3750 17500 15990 6500 2244 ...
## $ zipCode : int 90024 90024 90024 90024 90024 90024 90024 90024 90024 90024 ...
## $ bedrooms : int 1 0 2 2 0 2 4 4 2 1 ...
## $ bathrooms : num 1 1 2 3 1 1 4 4 2 1 ...
## $ squareFootage: int 680 700 1220 1888 562 1000 2956 2701 1403 NA ...
## $ propertyType : chr "Apartment" "Single Family" "Condo" "Condo" ...
## $ yearBuilt : int 1968 1939 2018 1980 1973 1947 1939 1950 1989 NA ...
## $ latitude : num 34.1 34.1 34.1 34.1 34.1 ...
## $ longitude : num -118 -118 -118 -118 -118 ...
```

```
colSums(is.na(data_clean))
```

```
##           price           zipCode      bedrooms      bathrooms squareFootage
##           0             0           0           2             273
##  propertyType    yearBuilt      latitude      longitude
##           0             795           0           0
```

```
data_clean$zipCode <- as.factor(data_clean$zipCode)
data_clean$propertyType <- as.factor(data_clean$propertyType)

## -----
## 3. Handle missing data and rare categories
## -----
# yearBuilt is missing for 31.6% of listings (795/2,519) -- too much
# structural missingness to keep or impute responsibly. Drop the variable.
data_clean2 <- data_clean %>% select(-yearBuilt)

# bathrooms (2 missing) and squareFootage (273 missing) are core variables
# that cannot be reasonably imputed -- remove those rows.
data_clean2 <- data_clean2 %>%
  filter(!is.na(bathrooms), !is.na(squareFootage))

nrow(data_clean2)
```

```
## [1] 2244
```

```
colSums(is.na(data_clean2))
```

```
##           price           zipCode      bedrooms      bathrooms squareFootage
##           0             0           0           0             0
##  propertyType    latitude      longitude
##           0             0           0
```

```
table(data_clean2$zipCode)
```

```
##
## 90019 90024 90025 90034 90035 90036 90048 90049 90064 90066 90069 90230 90403
##   185   212   256   215   190   132   178   130   100   125   249    48    89
## 90404 90405
##    58    77
```

```
# propertyType: "Multi-Family" has only 3 observations, too few to
# reliably estimate its effect -- remove and drop the empty level.
table(data_clean2$propertyType)
```

```
##
## Apartment      Condo  Multi-Family  Single Family      Townhouse
##       1087         393           3         544         217
```

```
data_clean3 <- data_clean2 %>%
  filter(propertyType != "Multi-Family") %>%
  droplevels()

nrow(data_clean3)
```

```
## [1] 2241
```

```
table(data_clean3$propertyType)
```

```
##
##      Apartment      Condo Single Family      Townhouse
##      1087          393          544          217
```

```
table(data_clean3$zipCode)
```

```
##
## 90019 90024 90025 90034 90035 90036 90048 90049 90064 90066 90069 90230 90403
##   185   212   255   215   190   131   178   130   100   124   249   48    89
## 90404 90405
##    58    77
```

```
## -----
## 4. Test latitude/longitude as an alternative to zipCode
## -----
```

```
data_clean3$zipCode <- as.factor(data_clean3$zipCode)
data_clean3$propertyType <- as.factor(data_clean3$propertyType)

full_model_geo <- lm(price ~ zipCode + bedrooms + bathrooms + squareFootage +
  propertyType + latitude + longitude,
  data = data_clean3)
summary(full_model_geo)
```

```
##
## Call:
## lm(formula = price ~ zipCode + bedrooms + bathrooms + squareFootage +
##      propertyType + latitude + longitude, data = data_clean3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17145.0  -1149.6   -68.7    808.6   10927.4
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.799e+06  5.624e+05  -3.199  0.00140 **
## zipCode90024    2.187e+02  4.856e+02   0.450  0.65246
## zipCode90025   -4.603e+02  5.021e+02  -0.917  0.35935
## zipCode90034    5.534e+02  3.541e+02   1.563  0.11819
## zipCode90035    3.458e+01  2.884e+02   0.120  0.90459
## zipCode90036    9.022e+02  2.773e+02   3.253  0.00116 **
## zipCode90048    9.325e+02  3.156e+02   2.955  0.00316 **
```

```
## zipCode90049          5.167e+02  6.119e+02  0.844  0.39855
## zipCode90064         -6.153e+02  4.608e+02 -1.335  0.18191
## zipCode90066          1.028e+03  4.969e+02  2.069  0.03863 *
## zipCode90069          2.118e+03  3.725e+02  5.685 1.48e-08 ***
## zipCode90230          8.627e+02  5.191e+02  1.662  0.09664 .
## zipCode90403          6.515e+02  7.071e+02  0.921  0.35701
## zipCode90404          6.333e+01  6.598e+02  0.096  0.92355
## zipCode90405          1.285e+03  6.523e+02  1.970  0.04900 *
## bedrooms             4.129e+02  6.441e+01  6.410 1.77e-10 ***
## bathrooms             1.573e+03  8.045e+01 19.552 < 2e-16 ***
## squareFootage        4.968e-01  4.665e-02 10.650 < 2e-16 ***
## propertyTypeCondo      8.537e+02  1.307e+02  6.533 7.98e-11 ***
## propertyTypeSingle Family 1.998e+03  1.258e+02 15.890 < 2e-16 ***
## propertyTypeTownhouse  2.470e+02  1.634e+02  1.512  0.13080
## latitude              1.578e+04  5.782e+03  2.730  0.00638 **
## longitude             -1.065e+04  4.234e+03 -2.515  0.01196 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2130 on 2218 degrees of freedom
## Multiple R-squared:  0.6517, Adjusted R-squared:  0.6483
## F-statistic: 188.7 on 22 and 2218 DF,  p-value: < 2.2e-16
```

```
vif(full_model_geo) # latitude/longitude show VIF = 11.4, 18.0 -- highly
```

```
##              GVIF Df GVIF^(1/(2*Df))
## zipCode      200.217190 14      1.208360
## bedrooms      3.083518  1      1.755995
## bathrooms      3.136101  1      1.770904
## squareFootage  1.378451  1      1.174074
## propertyType   1.373933  3      1.054373
## latitude      11.438279  1      3.382052
## longitude      17.980161  1      4.240302
```

```
# collinear with zipCode
```

```
reduced_model_nogeo <- lm(price ~ zipCode + bedrooms + bathrooms + squareFootage +
  propertyType,
  data = data_clean3)
```

```
# Partial F-test: does latitude/longitude add explanatory power beyond zipCode?
anova(reduced_model_nogeo, full_model_geo)
```

```
## Analysis of Variance Table
##
## Model 1: price ~ zipCode + bedrooms + bathrooms + squareFootage + propertyType
## Model 2: price ~ zipCode + bedrooms + bathrooms + squareFootage + propertyType +
## latitude + longitude
##   Res.Df      RSS Df Sum of Sq    F    Pr(>F)
## 1    2220 1.0122e+10
## 2    2218 1.0067e+10  2  55173644 6.0781 0.002331 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

# Statistically significant (p = 0.002) but a very small effect size
# (RSS drops by only ~0.55%) -- driven mainly by large sample size.
# Decision: drop latitude/longitude, keep zipCode alone for interpretability.

## -----
## 5. Generate near-final dataset (drop latitude/longitude)
## -----
final_clean <- data_clean3 %>%
  select(-latitude, -longitude)

str(final_clean)

## 'data.frame': 2241 obs. of 6 variables:
## $ price : int 2795 2400 7500 7295 2500 3750 17500 15990 6500 4325 ...
## $ zipCode : Factor w/ 15 levels "90019","90024",...: 2 2 2 2 2 2 2 2 2 ...
## $ bedrooms : int 1 0 2 2 0 2 4 4 2 1 ...
## $ bathrooms : num 1 1 2 3 1 1 4 4 2 1 ...
## $ squareFootage: int 680 700 1220 1888 562 1000 2956 2701 1403 650 ...
## $ propertyType : Factor w/ 4 levels "Apartment","Condo",...: 1 3 2 2 2 2 3 3 2 1 ...

nrow(final_clean)

## [1] 2241

colnames(final_clean)

## [1] "price" "zipCode" "bedrooms" "bathrooms"
## [5] "squareFootage" "propertyType"

colSums(is.na(final_clean))

## price zipCode bedrooms bathrooms squareFootage
## 0 0 0 0 0
## propertyType
## 0

table(final_clean$zipCode)

##
## 90019 90024 90025 90034 90035 90036 90048 90049 90064 90066 90069 90230 90403
## 185 212 255 215 190 131 178 130 100 124 249 48 89
## 90404 90405
## 58 77

table(final_clean$propertyType)

##
## Apartment Condo Single Family Townhouse
## 1087 393 544 217

```

```
## -----
## 6. Additional data-quality check: squareFootage / price-per-sqft
## -----
sum(final_clean$squareFootage == 0)

## [1] 1

final_clean[final_clean$squareFootage == 0, ]

##      price zipCode bedrooms bathrooms squareFootage propertyType
## 1535  3800   90048         1          1             0      Townhouse

final_clean$price_per_sqft <- final_clean$price / final_clean$squareFootage
summary(final_clean$price_per_sqft)

##      Min. 1st Qu.  Median      Mean 3rd Qu.     Max.
## 0.09545 3.15534 3.79197      Inf 4.77587      Inf

sum(is.infinite(final_clean$price_per_sqft))

## [1] 1

sum(final_clean$price_per_sqft < 1, na.rm = TRUE)

## [1] 26

sum(final_clean$price_per_sqft > 20, na.rm = TRUE)

## [1] 3

# Remove: squareFootage = 0 (1 obs), price/sqft < $1 (26 obs) -- implausible,
# consistent with data entry errors (same threshold used in Mini Assignment 2)
final_clean_low <- final_clean %>%
  filter(squareFootage > 0, price_per_sqft >= 1)
nrow(final_clean_low)

## [1] 2214

# Inspect the high end before deciding
final_clean_low %>%
  filter(price_per_sqft > 20) %>%
  select(price, squareFootage, price_per_sqft, zipCode, propertyType)

##      price squareFootage price_per_sqft zipCode propertyType
## 1   4965           213      23.30986   90024   Apartment
## 2   1607            78      20.60256   90024   Apartment
```

```

# 2 listings with implausibly small squareFootage relative to price
# (e.g., a 78 sq ft "apartment") -- remove as data entry errors.

final_clean_final <- final_clean_low %>%
  filter(price_per_sqft <= 20) %>%
  select(-price_per_sqft)

nrow(final_clean_final)  # 2,212 observations -- final analytic sample

## [1] 2212

str(final_clean_final)

## 'data.frame':    2212 obs. of  6 variables:
##  $ price          : int  2795 2400 7500 7295 2500 3750 17500 15990 6500 4325 ...
##  $ zipCode        : Factor w/ 15 levels "90019","90024",...: 2 2 2 2 2 2 2 2 2 ...
##  $ bedrooms       : int   1 0 2 2 0 2 4 4 2 1 ...
##  $ bathrooms      : num   1 1 2 3 1 1 4 4 2 1 ...
##  $ squareFootage: int   680 700 1220 1888 562 1000 2956 2701 1403 650 ...
##  $ propertyType   : Factor w/ 4 levels "Apartment","Condo",...: 1 3 2 2 2 2 3 3 2 1 ...

## =====
## DATA DESCRIPTION -- tables and figures reported in the report
## =====

final_clean_final$zipCode <- as.factor(final_clean_final$zipCode)
final_clean_final$propertyType <- as.factor(final_clean_final$propertyType)

cont_vars <- final_clean_final[, c("price", "bedrooms", "bathrooms", "squareFootage")]

## ---- Summary statistics (reported in main report) ----
summary_table <- data.frame(
  Variable = names(cont_vars),
  Min      = sapply(cont_vars, min),
  Q1       = sapply(cont_vars, quantile, 0.25),
  Median   = sapply(cont_vars, median),
  Mean     = sapply(cont_vars, mean),
  Q3       = sapply(cont_vars, quantile, 0.75),
  Max      = sapply(cont_vars, max),
  SD       = sapply(cont_vars, sd)
)
print(summary_table, row.names = FALSE)

##      Variable Min   Q1 Median      Mean      Q3   Max      SD
##      price  750 2650 3600.0 4881.223779 5661.25 20000 3606.3165249
##      bedrooms    0    1    2.0   1.882007    3.00    14    1.2198798
##      bathrooms    1    1    1.5   1.808996    2.00     8    0.9924069
##      squareFootage  90  700 1015.5 1192.845841 1450.25  5505  723.8384336

## ---- Categorical distributions (Appendix Table A1) ----
table(final_clean_final$zipCode)

```

```
##
## 90019 90024 90025 90034 90035 90036 90048 90049 90064 90066 90069 90230 90403
## 182 206 254 212 186 130 176 128 100 122 249 48 89
## 90404 90405
## 58 72
```

```
table(final_clean_final$propertyType)
```

```
##
## Apartment Condo Single Family Townhouse
## 1076 388 538 210
```

```
## ---- Mean price by zip code (Appendix Table A2) ----
tapply(final_clean_final$price, final_clean_final$zipCode, mean)
```

```
## 90019 90024 90025 90034 90035 90036 90048 90049
## 3726.352 5067.971 4145.709 3471.623 4823.581 4097.154 5435.307 7087.344
## 90064 90066 90069 90230 90403 90404 90405
## 4698.190 4862.041 6889.498 4183.562 4727.472 3849.103 5127.611
```

```
## ---- Correlation matrix (Appendix Table A3) ----
cor(cont_vars)
```

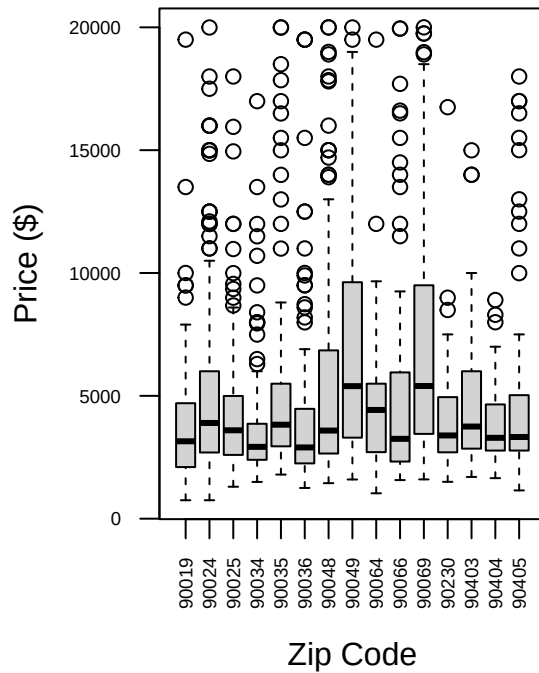
```
## price bedrooms bathrooms squareFootage
## price 1.0000000 0.6411132 0.7039040 0.8532168
## bedrooms 0.6411132 1.0000000 0.8040380 0.7639440
## bathrooms 0.7039040 0.8040380 1.0000000 0.7969624
## squareFootage 0.8532168 0.7639440 0.7969624 1.0000000
```

```
par(mfrow = c(1, 2))

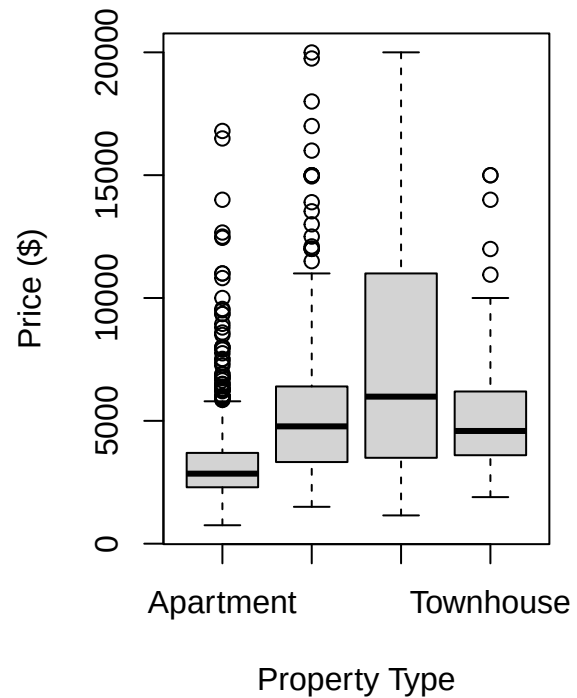
boxplot(price ~ zipCode, data = final_clean_final,
  main = "Rental Price by Zip Code",
  xlab = "Zip Code", ylab = "Price ($)",
  las = 2, cex.axis = 0.6)

boxplot(price ~ propertyType, data = final_clean_final,
  main = "Rental Price by Property Type",
  xlab = "Property Type", ylab = "Price ($)")
```

Rental Price by Zip Code

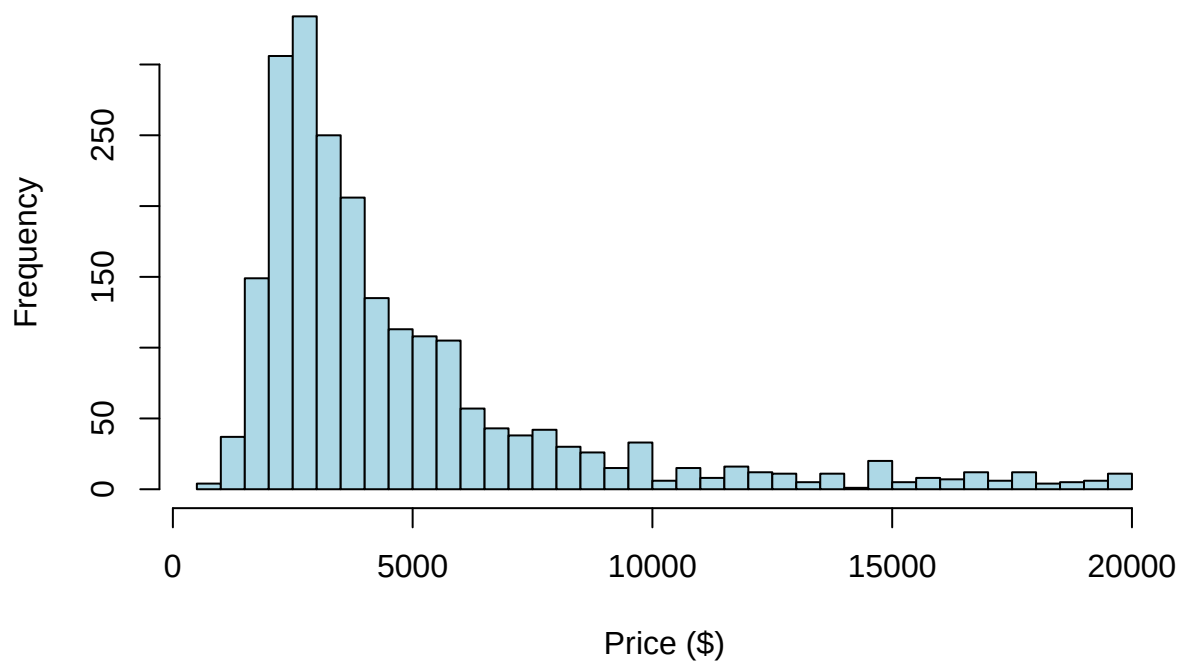


Rental Price by Property Type



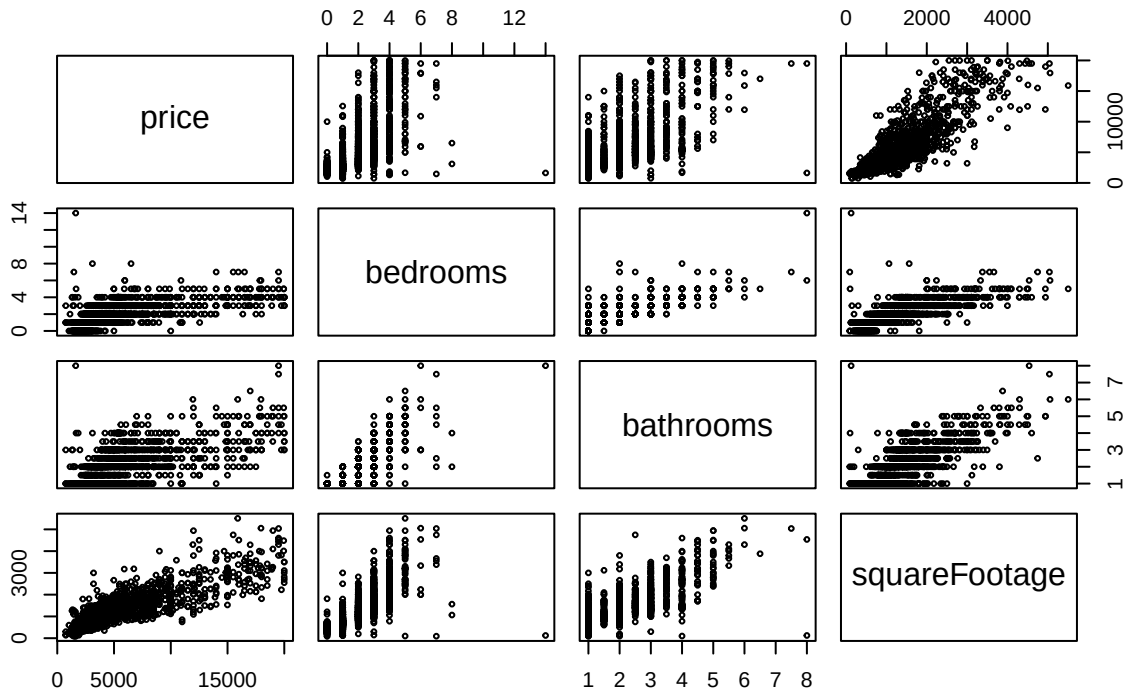
```
par(mfrow = c(1, 1))
hist(final_clean_final$price,
     main = "Distribution of Monthly Rental Price",
     xlab = "Price ($)", col = "lightblue", breaks = 40)
```

Distribution of Monthly Rental Price



```
pairs(~ price + bedrooms + bathrooms + squareFootage, data = final_clean_final,  
      main = "Scatter Plot Matrix", pch = 1, cex = 0.5)
```

Scatter Plot Matrix



```
my_data1 <- final_clean_final

my_data1_clean <- my_data1 %>%
  filter(!(bedrooms == 14 & bathrooms == 8 & squareFootage == 125))
```

After applying the stated cleaning criteria, one additional listing was removed because it reported 14 bedrooms and 8 bathrooms within 125 square feet, a physically implausible combination that could not be verified. The resulting analytic sample contained 2211 listings.

```
model_full <- lm(price ~ zipCode + bedrooms + bathrooms + squareFootage + propertyType, data = my_data1_clean)
summary(model_full)
```

```
##
## Call:
## lm(formula = price ~ zipCode + bedrooms + bathrooms + squareFootage +
##     propertyType, data = my_data1_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8662.2  -912.7   -99.7    709.5   9050.5
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.502e+03  1.442e+02 -10.415  < 2e-16 ***
## zipCode90024    1.353e+03  1.747e+02   7.748  1.42e-14 ***
```

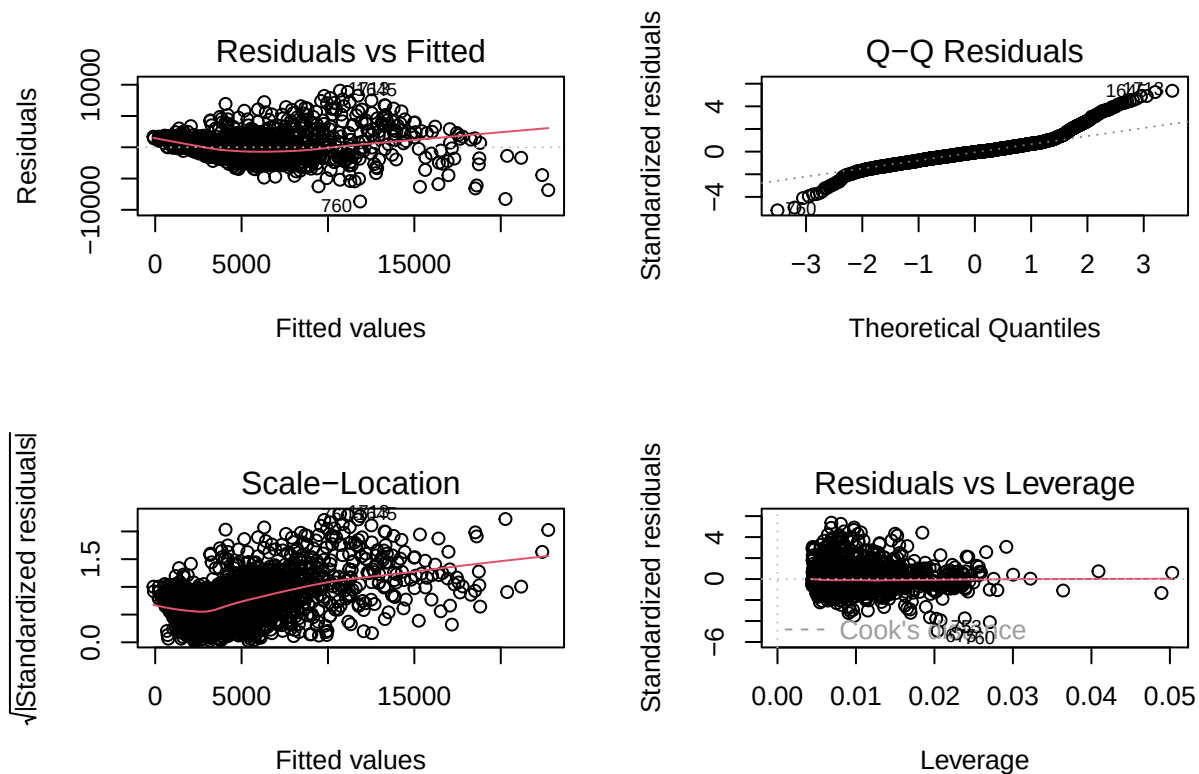
```
## zipCode90025          6.243e+02  1.649e+02   3.787 0.000157 ***
## zipCode90034          9.405e+02  1.718e+02   5.475 4.88e-08 ***
## zipCode90035          3.464e+02  1.768e+02   1.960 0.050146 .
## zipCode90036          8.894e+02  1.967e+02   4.521 6.47e-06 ***
## zipCode90048          1.456e+03  1.797e+02   8.103 8.79e-16 ***
## zipCode90049          1.308e+03  1.988e+02   6.577 5.98e-11 ***
## zipCode90064          5.222e+02  2.138e+02   2.443 0.014660 *
## zipCode90066          1.178e+03  1.985e+02   5.935 3.40e-09 ***
## zipCode90069          2.651e+03  1.673e+02  15.848 < 2e-16 ***
## zipCode90230          6.578e+02  2.748e+02   2.394 0.016768 *
## zipCode90403          1.652e+03  2.202e+02   7.500 9.21e-14 ***
## zipCode90404          1.121e+03  2.556e+02   4.385 1.21e-05 ***
## zipCode90405          2.253e+03  2.359e+02   9.548 < 2e-16 ***
## bedrooms             -8.567e+01  5.724e+01  -1.497 0.134608
## bathrooms            4.557e+02  7.154e+01   6.371 2.29e-10 ***
## squareFootage        3.574e+00  9.977e-02  35.821 < 2e-16 ***
## propertyTypeCondo     4.926e+02  1.049e+02   4.697 2.80e-06 ***
## propertyTypeSingle Family 9.654e+02  1.043e+02   9.253 < 2e-16 ***
## propertyTypeTownhouse  5.645e+01  1.316e+02   0.429 0.667995
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1690 on 2190 degrees of freedom
## Multiple R-squared:  0.7824, Adjusted R-squared:  0.7804
## F-statistic: 393.8 on 20 and 2190 DF, p-value: < 2.2e-16
```

ZIP code 90019 and Apartment serve as the reference categories for the two categorical predictors.

The fitted model explains approximately 78.24% of the variability in monthly rental prices, with an adjusted R^2 of 0.7804. The overall regression is statistically significant, $F(20,2190)=393.8$, $p<0.001$, indicating that the predictors collectively provide significant explanatory power for monthly rent.

Bathrooms and square footage are statistically significant predictors of rent. Holding the other variables constant, one additional bathroom is associated with an estimated 456 dollar increase in monthly rent, while one additional square foot is associated with an estimated \$3.57 increase. Bedrooms are not statistically significant at the 5% level.

```
par(mfrow = c(2, 2))
plot(model_full)
```



```
par(mfrow = c(1, 1))
```

The diagnostic plots indicate that the original full model does not fully satisfy the regression assumptions. The Residuals vs. Fitted plot shows a curved pattern, suggesting that the linear specification may be inadequate. The Normal Q-Q plot shows substantial departures from normality in both tails. The Scale-Location plot provides strong evidence of heteroscedasticity, with residual variability increasing as fitted rent increases. The Residuals vs. Leverage plot also identifies several potentially unusual or influential observations. Therefore, a transformation of the response variable should be considered before variable selection.

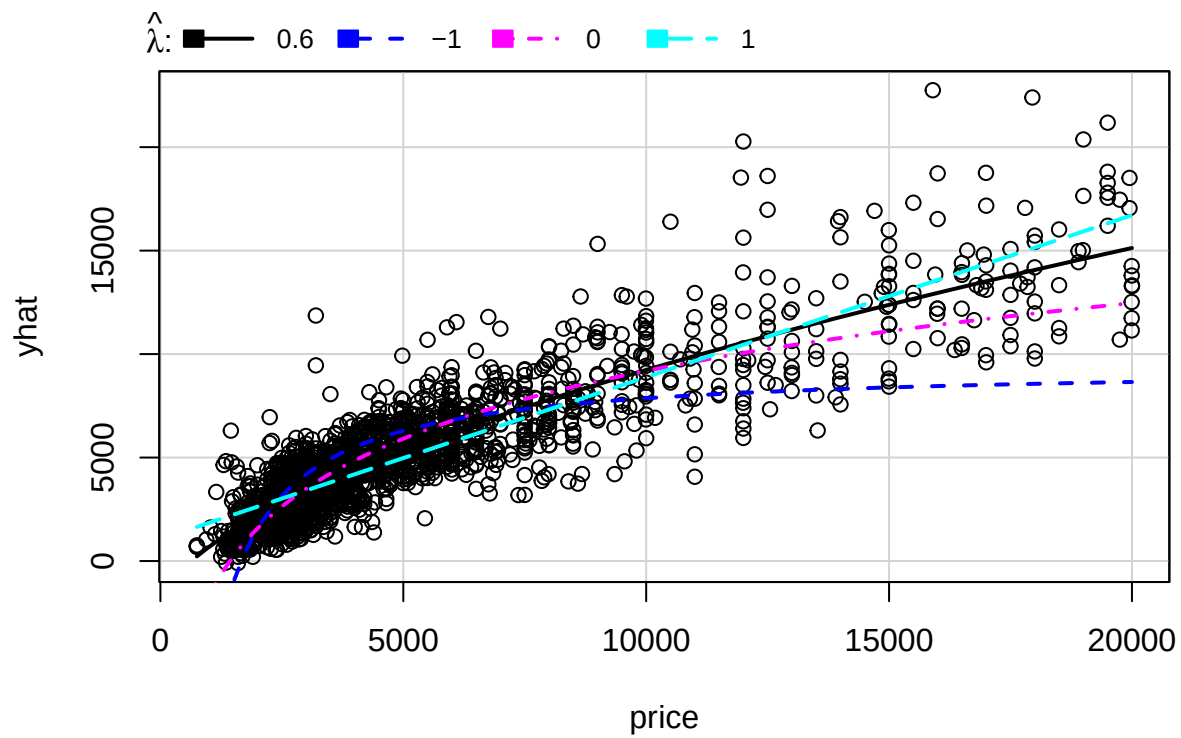
```
vif(model_full)
```

```
##           GVIF Df GVIF^(1/(2*Df))
## zipCode    1.315736 14      1.009848
## bedrooms   3.605927  1      1.898928
## bathrooms  3.833188  1      1.957853
## squareFootage 4.033672 1      2.008400
## propertyType 1.490916  3      1.068831
```

There is no evidence of serious multicollinearity among the predictors. All adjusted GVIF values are below 2, indicating that the predictors do not exhibit problematic linear dependence.

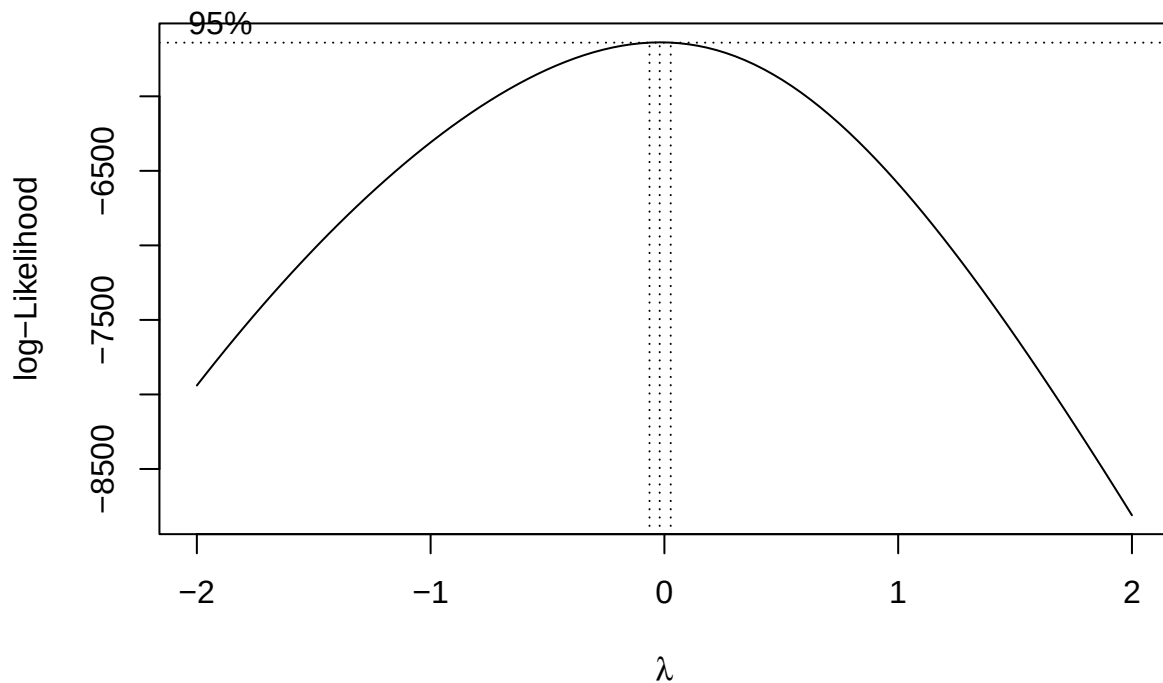
Model transformation

```
inverseResponsePlot(model_full)
```



##	lambda	RSS
## 1	0.600028	4553209370
## 2	-1.000000	9988238870
## 3	0.000000	5417212788
## 4	1.000000	4893527865

```
MASS::boxcox(model_full)
```



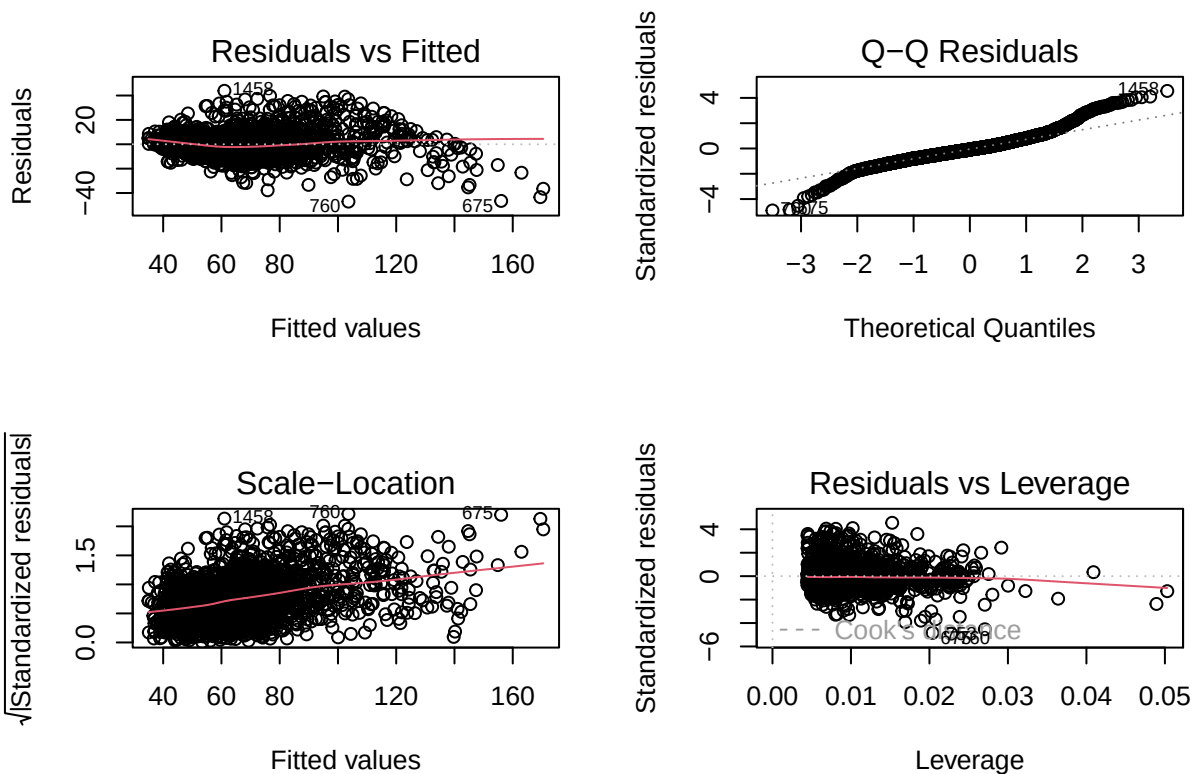
The inverse response plot estimated lambda approximately to be 0.60, suggesting a transformation close to the square root, whereas the Box-Cox profile likelihood was maximized near lambda = 0, suggesting a logarithmic transformation. Therefore, both transformations were evaluated.

```
model_sqrt <- lm(sqrt(price) ~ zipCode + bedrooms + bathrooms + squareFootage + propertyType, data = my_data1_clean)
summary(model_sqrt)
```

```
##
## Call:
## lm(formula = sqrt(price) ~ zipCode + bedrooms + bathrooms + squareFootage +
##     propertyType, data = my_data1_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -47.030  -5.469   -0.807    4.512   43.855
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.583e+01  8.285e-01  31.178 < 2e-16 ***
## zipCode90024   9.345e+00  1.003e+00   9.314 < 2e-16 ***
## zipCode90025   4.962e+00  9.470e-01   5.240 1.76e-07 ***
## zipCode90034   6.116e+00  9.868e-01   6.198 6.80e-10 ***
## zipCode90035   3.253e+00  1.015e+00   3.204 0.001377 **
## zipCode90036   5.723e+00  1.130e+00   5.065 4.44e-07 ***
## zipCode90048   9.454e+00  1.032e+00   9.157 < 2e-16 ***
```

```
## zipCode90049          9.174e+00  1.142e+00   8.034 1.53e-15 ***
## zipCode90064          5.110e+00  1.228e+00   4.161 3.29e-05 ***
## zipCode90066          7.512e+00  1.140e+00   6.588 5.58e-11 ***
## zipCode90069          1.774e+01  9.608e-01  18.461 < 2e-16 ***
## zipCode90230          4.686e+00  1.579e+00   2.968 0.003026 **
## zipCode90403          1.155e+01  1.265e+00   9.129 < 2e-16 ***
## zipCode90404          8.072e+00  1.468e+00   5.498 4.30e-08 ***
## zipCode90405          1.371e+01  1.355e+00  10.118 < 2e-16 ***
## bedrooms             1.235e+00  3.288e-01   3.755 0.000178 ***
## bathrooms             2.394e+00  4.109e-01   5.825 6.54e-09 ***
## squareFootage         1.955e-02  5.731e-04  34.115 < 2e-16 ***
## propertyTypeCondo      5.585e+00  6.024e-01   9.272 < 2e-16 ***
## propertyTypeSingle Family 6.310e+00  5.993e-01  10.528 < 2e-16 ***
## propertyTypeTownhouse  3.122e+00  7.559e-01   4.130 3.77e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.707 on 2190 degrees of freedom
## Multiple R-squared:  0.8029, Adjusted R-squared:  0.8011
## F-statistic: 446.2 on 20 and 2190 DF, p-value: < 2.2e-16
```

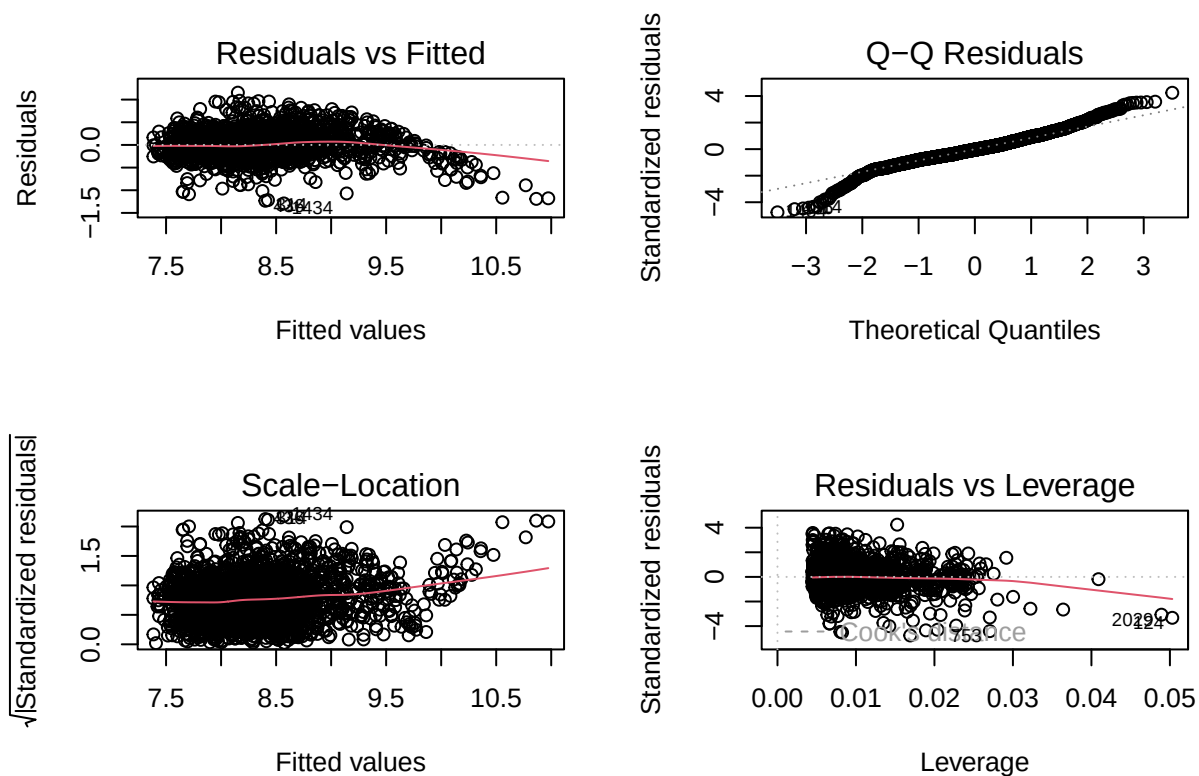
```
par(mfrow = c(2, 2))
plot(model_sqrt)
```



```
model_log <- lm(log(price) ~ zipCode + bedrooms + bathrooms + squareFootage + propertyType, data = my_data)
summary(model_log)
```

```
##
## Call:
## lm(formula = log(price) ~ zipCode + bedrooms + bathrooms + squareFootage +
##     propertyType, data = my_data1_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.29136 -0.15744 -0.00894  0.15564  1.15302
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.177e+00  2.336e-02 307.219 < 2e-16 ***
## zipCode90024    2.785e-01  2.829e-02   9.846 < 2e-16 ***
## zipCode90025    1.709e-01  2.670e-02   6.402 1.87e-10 ***
## zipCode90034    1.831e-01  2.782e-02   6.582 5.81e-11 ***
## zipCode90035    1.305e-01  2.863e-02   4.559 5.42e-06 ***
## zipCode90036    1.631e-01  3.186e-02   5.120 3.33e-07 ***
## zipCode90048    2.733e-01  2.911e-02   9.388 < 2e-16 ***
## zipCode90049    2.838e-01  3.220e-02   8.814 < 2e-16 ***
## zipCode90064    1.933e-01  3.462e-02   5.583 2.66e-08 ***
## zipCode90066    2.137e-01  3.215e-02   6.648 3.75e-11 ***
## zipCode90069    5.161e-01  2.709e-02  19.050 < 2e-16 ***
## zipCode90230    1.547e-01  4.451e-02   3.475 0.00052 ***
## zipCode90403    3.544e-01  3.567e-02   9.936 < 2e-16 ***
## zipCode90404    2.601e-01  4.140e-02   6.284 3.97e-10 ***
## zipCode90405    3.710e-01  3.821e-02   9.709 < 2e-16 ***
## bedrooms        8.331e-02  9.270e-03   8.987 < 2e-16 ***
## bathrooms        4.797e-02  1.159e-02   4.140 3.60e-05 ***
## squareFootage    4.591e-04  1.616e-05  28.411 < 2e-16 ***
## propertyTypeCondo 2.170e-01  1.698e-02  12.774 < 2e-16 ***
## propertyTypeSingle Family 1.720e-01  1.690e-02  10.179 < 2e-16 ***
## propertyTypeTownhouse 1.573e-01  2.131e-02   7.381 2.22e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2737 on 2190 degrees of freedom
## Multiple R-squared:  0.7853, Adjusted R-squared:  0.7833
## F-statistic: 400.5 on 20 and 2190 DF, p-value: < 2.2e-16

par(mfrow = c(2, 2))
plot(model_log)
```



```
par(mfrow = c(1, 1))
```

Although neither transformation completely eliminated departures from normality or constant variance, the logarithmic model showed a more stable residual spread and slightly less curvature in the residual plots. In addition, the Box-Cox analysis suggested a transformation parameter close to zero, which corresponds to a logarithmic transformation. Therefore, log-transformed monthly rent was selected as the response variable.

As the log transformation on the response variable isn't satisfying, separated residual plots were performed on individual continuous variables to assist further transformation on explanatory variables.

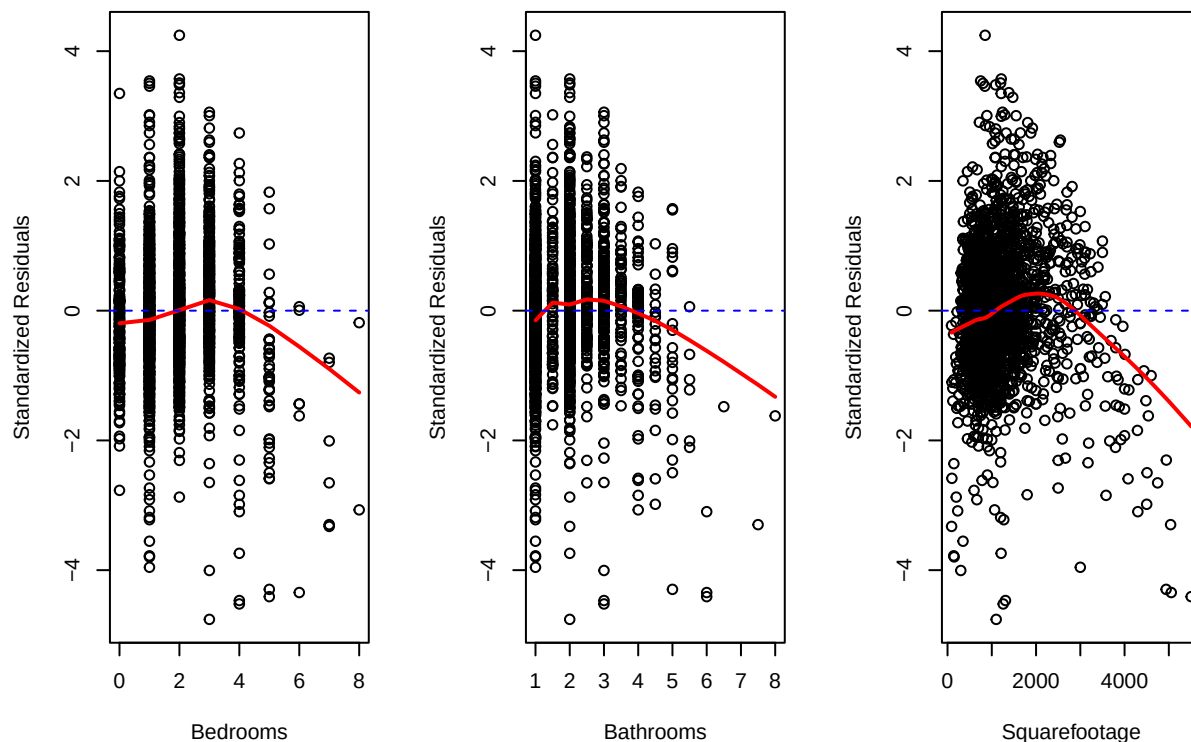
```
standardized_residuais <- rstandard(model_log)
```

```
par(mfrow = c(1, 3))
```

```
plot(my_data1_clean$bedrooms, standardized_residuais, xlab = "Bedrooms", ylab = "Standardized Residuals")
lines(lowess(my_data1_clean$bedrooms, standardized_residuais), col = "red", lwd = 2)
abline(h = 0, col = "blue", lty = 2)
```

```
plot(my_data1_clean$bathrooms, standardized_residuais, xlab = "Bathrooms", ylab = "Standardized Residuals")
lines(lowess(my_data1_clean$bathrooms, standardized_residuais), col = "red", lwd = 2)
abline(h = 0, col = "blue", lty = 2)
```

```
plot(my_data1_clean$squareFootage, standardized_residuais, xlab = "Squarefootage", ylab = "Standardized Residuals")
lines(lowess(my_data1_clean$squareFootage, standardized_residuais), col = "red", lwd = 2)
abline(h = 0, col = "blue", lty = 2)
```



```
par(mfrow = c(1, 1))
```

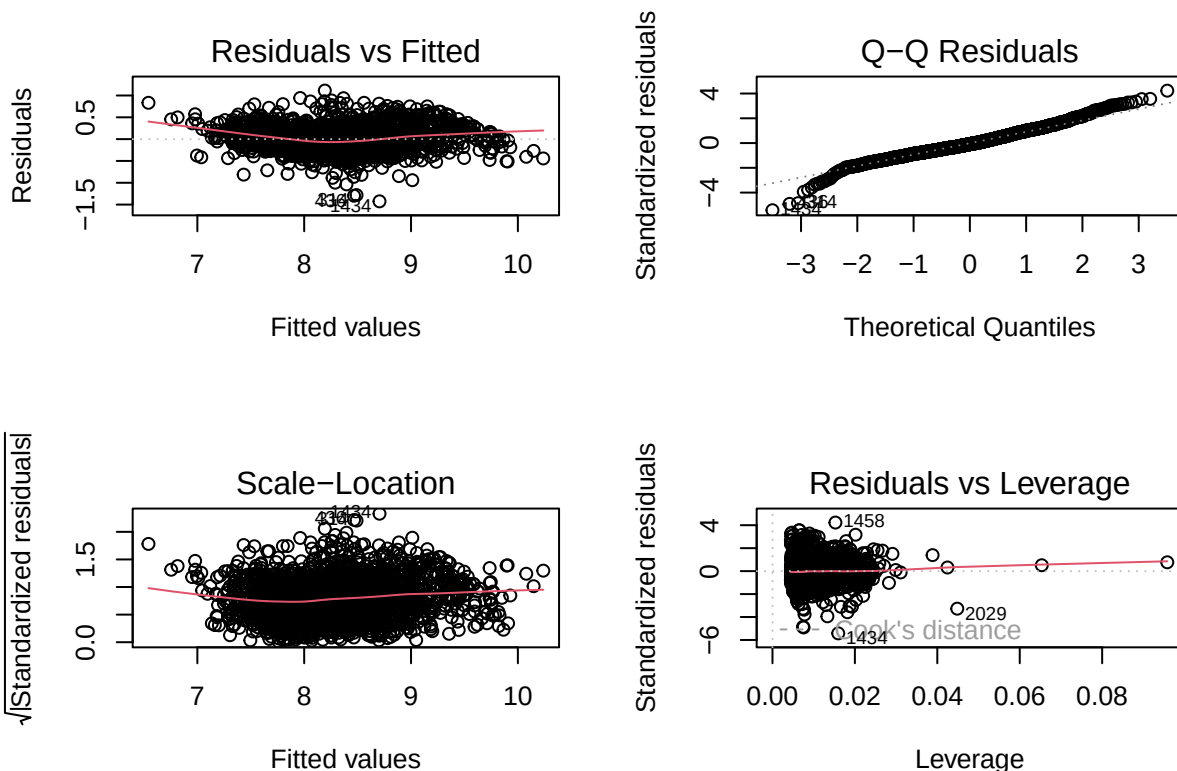
All three individual variables show some degrees of pattern. But, relatively speaking, the square footage's downward pattern is supported by larger sample observations. So transformation on square footage is considered first.

```
model_log_logsf <- lm(log(price) ~ zipCode + bedrooms + bathrooms + log(squareFootage) + propertyType, data = my_data1_clean)
summary(model_log_logsf)
```

```
##
## Call:
## lm(formula = log(price) ~ zipCode + bedrooms + bathrooms + log(squareFootage) +
##     propertyType, data = my_data1_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.42576 -0.16685 -0.02243  0.16091  1.11266
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.639965   0.113967  31.939 < 2e-16 ***
## zipCode90024   0.309183   0.027346  11.306 < 2e-16 ***
## zipCode90025   0.153542   0.025863   5.937 3.37e-09 ***
## zipCode90034   0.182885   0.026943   6.788 1.46e-11 ***
## zipCode90035   0.096357   0.027783   3.468 0.000534 ***
```

```
## zipCode90036      0.193474  0.030773  6.287 3.89e-10 ***
## zipCode90048      0.253168  0.028219  8.972 < 2e-16 ***
## zipCode90049      0.317562  0.031008 10.241 < 2e-16 ***
## zipCode90064      0.153989  0.033460  4.602 4.42e-06 ***
## zipCode90066      0.215803  0.031133  6.932 5.45e-12 ***
## zipCode90069      0.487578  0.026318 18.526 < 2e-16 ***
## zipCode90230      0.143567  0.043097  3.331 0.000879 ***
## zipCode90403      0.352486  0.034541 10.205 < 2e-16 ***
## zipCode90404      0.260867  0.040090  6.507 9.47e-11 ***
## zipCode90405      0.409614  0.036997 11.072 < 2e-16 ***
## bedrooms          0.039107  0.009471  4.129 3.77e-05 ***
## bathrooms         0.110868  0.010326 10.736 < 2e-16 ***
## log(squareFootage) 0.586240  0.018484 31.715 < 2e-16 ***
## propertyTypeCondo  0.165520  0.016659  9.936 < 2e-16 ***
## propertyTypeSingle Family 0.210970  0.015991 13.193 < 2e-16 ***
## propertyTypeTownhouse 0.112708  0.020732  5.436 6.04e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2651 on 2190 degrees of freedom
## Multiple R-squared:  0.7986, Adjusted R-squared:  0.7968
## F-statistic: 434.3 on 20 and 2190 DF, p-value: < 2.2e-16
```

```
par(mfrow = c(2, 2))
plot(model_log_logsf)
```



```

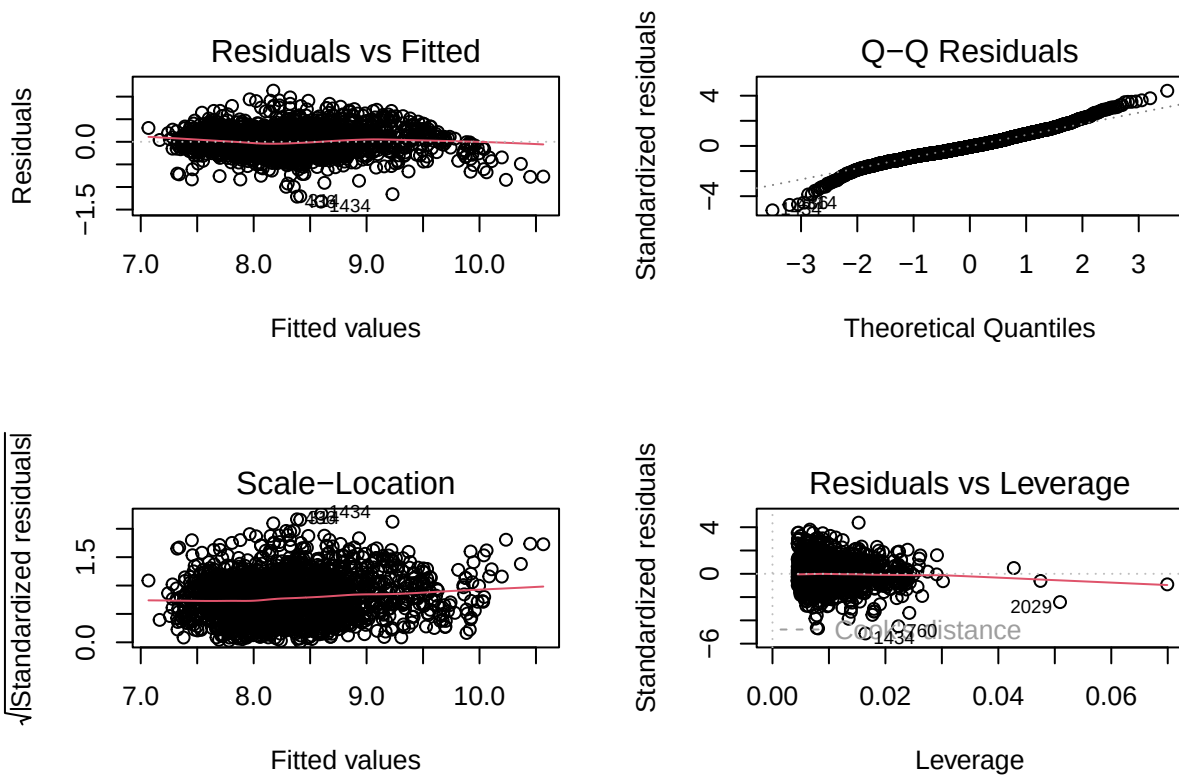
par(mfrow = c(1, 1))

model_log_sqrtsf <- lm(log(price) ~ zipCode + bedrooms + bathrooms + sqrt(squareFootage) + propertyType
summary(model_log_sqrtsf)

##
## Call:
## lm(formula = log(price) ~ zipCode + bedrooms + bathrooms + sqrt(squareFootage) +
##     propertyType, data = my_data1_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.31975 -0.15588 -0.01488  0.15178  1.13186
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.468912   0.030638 211.137 < 2e-16 ***
## zipCode90024      0.281841   0.026827  10.506 < 2e-16 ***
## zipCode90025      0.159311   0.025339   6.287 3.89e-10 ***
## zipCode90034      0.186338   0.026402   7.058 2.26e-12 ***
## zipCode90035      0.104759   0.027201   3.851 0.000121 ***
## zipCode90036      0.168255   0.030196   5.572 2.83e-08 ***
## zipCode90048      0.256567   0.027641   9.282 < 2e-16 ***
## zipCode90049      0.282193   0.030492   9.255 < 2e-16 ***
## zipCode90064      0.182332   0.032819   5.556 3.10e-08 ***
## zipCode90066      0.214550   0.030507   7.033 2.70e-12 ***
## zipCode90069      0.484827   0.025781  18.806 < 2e-16 ***
## zipCode90230      0.151204   0.042232   3.580 0.000351 ***
## zipCode90403      0.347223   0.033850  10.258 < 2e-16 ***
## zipCode90404      0.261255   0.039283   6.651 3.68e-11 ***
## zipCode90405      0.393255   0.036250  10.848 < 2e-16 ***
## bedrooms          0.037933   0.009210   4.119 3.95e-05 ***
## bathrooms          0.056444   0.010632   5.309 1.21e-07 ***
## sqrt(squareFootage) 0.040450   0.001199  33.741 < 2e-16 ***
## propertyTypeCondo  0.178039   0.016240  10.963 < 2e-16 ***
## propertyTypeSingle Family 0.168628   0.015924  10.589 < 2e-16 ***
## propertyTypeTownhouse 0.124919   0.020275   6.161 8.57e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2597 on 2190 degrees of freedom
## Multiple R-squared:  0.8067, Adjusted R-squared:  0.8049
## F-statistic: 456.9 on 20 and 2190 DF, p-value: < 2.2e-16

par(mfrow = c(2, 2))
plot(model_log_sqrtsf)

```



```
par(mfrow = c(1, 1))
```

```
AIC(model_log_logsf, model_log_sqrtsf)
```

```
##           df      AIC
## model_log_logsf 22 425.8573
## model_log_sqrtsf 22 335.9945
```

```
BIC(model_log_logsf, model_log_sqrtsf)
```

```
##           df      BIC
## model_log_logsf 22 551.2837
## model_log_sqrtsf 22 461.4209
```

```
summary(model_log_logsf)$adj.r.squared
```

```
## [1] 0.7968088
```

```
summary(model_log_sqrtsf)$adj.r.squared
```

```
## [1] 0.8049017
```

By observing the plots, both transformation methods show similar patterns and no distinct differences. Thus, comparison between AIC and BIC is implemented. The result shows that the square-root model had substantially lower AIC and BIC values and diagnostics comparable to the logarithmic model. Therefore, we select square-root transformation as the best available model within the transformation methods considered. Some remaining departures from linearity, constant variance, and normality are acknowledged as limitations.

```
model_full_trans <- lm(log(price) ~ zipCode + bedrooms + bathrooms + sqrt(squareFootage) + propertyType)

model_backward <- step(model_full_trans, direction = "backward", trace = 0)

model_null <- lm(log(price) ~ 1, data = my_data1_clean)
model_forward <- step(model_null, scope = formula(model_full_trans), direction = "forward", trace = 0)

formula(model_forward)
```

```
## log(price) ~ sqrt(squareFootage) + zipCode + propertyType + bathrooms +
##      bedrooms
```

```
formula(model_backward)
```

```
## log(price) ~ zipCode + bedrooms + bathrooms + sqrt(squareFootage) +
##      propertyType
```

Both forward and backward stepwise selection based on AIC resulted in the same model as the transformed full model. Therefore, the stepwise procedures provided no evidence that any predictor should be removed.

```
standardized_residuals <- rstandard(model_full_trans)
leverage <- hatvalues(model_full_trans)
cooks_distance <- cooks.distance(model_full_trans)

sum(abs(standardized_residuals) > 3)
```

```
## [1] 28
```

```
sum(leverage > 2 * length(coef(model_full_trans)) / nrow(my_data1_clean))
```

```
## [1] 103
```

```
sum(cooks_distance > 4 / nrow(my_data1_clean))
```

```
## [1] 110
```

```
max(cooks_distance)
```

```
## [1] 0.0221118
```

Using conventional screening thresholds, 28 observations had absolute standardized residuals greater than 3, 103 exceeded the leverage threshold of $2p/n$, and 110 exceeded the Cook's distance threshold of $4/n$. However, the maximum Cook's distance was only 0.022, indicating that no single observation exerted exceptionally strong influence on the fitted model. Therefore, these observations were retained